DREW T. ERNST

Frontend Architect | Full-Stack Engineer | Data Analyst

Quincy, IL • (217) 653-9445 • dte.solutions.llc@gmail.com GitHub • Website • LinkedIn

EXECUTIVE PROFILE

Elite software architect bridging advanced engineering with strategic communication. Dual degrees in Public Relations and Marketing enable translation of complex algorithmic logic into intuitive user experiences. Specialized in engineering production-grade systems from HIPAA-compliant clinical engines to Behavioral AI fin-tech platforms, with proven ability to transform technical velocity into measurable business value.

TECHNICAL EXPERTISE

Languages:	TypeScript, JavaScript (ES6+), Python, Java, C#, SQL (PostgreSQL), NoSQL (MongoDB)
Frameworks:	React 19, Next.js, Node.js, FastAPI, .NET Core
Cloud & DevOps:	AWS (S3, EC2), Docker, CI/CD (GitHub Actions), Vercel
AI Integration:	GPT-4o, Claude 3.5, Behavioral Modeling, Long-Term Memory Systems
Methodology:	SDLC, Object-Oriented Design, Agile/Scrum, High-Fidelity UI/UX Design

ENGINEERING LEADERSHIP

Founder & Principal Consultant | DTE Solutions LLC

2025 – Present • Tech Stack: TypeScript, Next.js, Node.js, C#, .NET Core, AWS, Docker

- Established independent consultancy engineering specialized analytical tools using Next.js for frontend and Node.js for backend data orchestration
- Designed high-performance API integration road map for automotive client, navigating complex data-sensitivity constraints to architect secure inventory management interface
- Orchestrated full product life cycle utilizing Docker for containerization and AWS (EC2/S3) for production deployments with GitHub Actions CI/CD

Founder & Principal Engineer | NestLegacy

2026 • Digital Trust Layer for $84T Wealth Transfer • Tech Stack: Next.js, TypeScript, GPT-4o, Tailwind CSS

- Engineered AI-driven behavioral lead qualification engine utilizing GPT-4o via OpenAI API, integrated into Next.js application
- Designed interactive longevity and tax-exposure calculators using TypeScript, delivering immediate mathematical value through high-fidelity UI components
- Developed modular hub-style architecture deployed on Vercel ensuring global scalability and data integrity across enterprise branches

Lead Full-Stack Developer | Pulse (Behavioral Finance AI)

2026 • Tech Stack: React 19, FastAPI (Python), PostgreSQL, MongoDB, AWS S3, Recharts

- Engineered "Nova," an evolving AI consciousness using GPT-4o implementing behavioral mirroring to correlate PostgreSQL financial data with user stress indices
- Designed relational database schemas in PostgreSQL maintaining strict ACID compliance for financial transactional data with detailed behavioral tracking
- Architected multi-tier system utilizing React 19 and FastAPI asynchronous backend achieving sub-millisecond data delivery to Recharts visualization
- Integrated AWS S3 receipt system utilizing encrypted pre-signed URL protocols ensuring audit-ready data persistence and uncompromising user privacy

Lead Software Engineer | Marion County DDSD (Missouri DMH)

2026 • HIPAA-Compliant Clinical Documentation Engine • Tech Stack: JavaScript, Java, OOD, CSS3, HTML5

- Developed intelligent phrasing engine in vanilla JavaScript applying Object-Oriented Design to translate complex clinical requirements into intuitive UI patterns
- Reduced documentation revision cycles by designing custom state management system ensuring 100% adherence to Missouri DMH state audit requirements
- Engineered in-memory processing architecture satisfying stringent HIPAA security standards without compromising system speed or data integrity

Full-Stack Engineer | Caddie & Fluff Ecosystem

2025 – 2026 • Biomechanical Intelligence & Performance Tracking • Tech Stack: React, TypeScript, Node.js, MongoDB, Firebase

- Engineered computer vision engine using Python and OpenCV to analyze skeletal alignment, streaming real-time kinematic feedback to React mobile interface
- Developed "Live Skins" synchronization engine using Node.js, Socket.io, and MongoDB providing real-time competitive precision for multiple users
- Leveraged Firebase (Auth & Firestore) as primary real-time database managing user state, GPS coordinates, and tactical course advice synchronization

EDUCATION & TRAINING

Advanced Engineering Mentorship | David Bragg (Senior Software Engineer) | 2025 – 2026

Intensive 1-on-1 technical mentorship focused on enterprise-grade software architecture, advanced OOD principles, and production lifecycle management

Frontend Simplified Bootcamp | Production-Grade React & Software Architecture | 2025

Comprehensive immersion in modern frontend ecosystems, focusing on state management complexity and elite UI implementation

B.S. Communication (Public Relations Focus) | Quincy University | GPA: 3.8

B.S. Marketing | Culver-Stockton College | GPA: 3.8